

2. The UK has been generating more energy using wind power over recent years.



- (a) Circle **one** renewable energy source below.

[1]

coal

gas

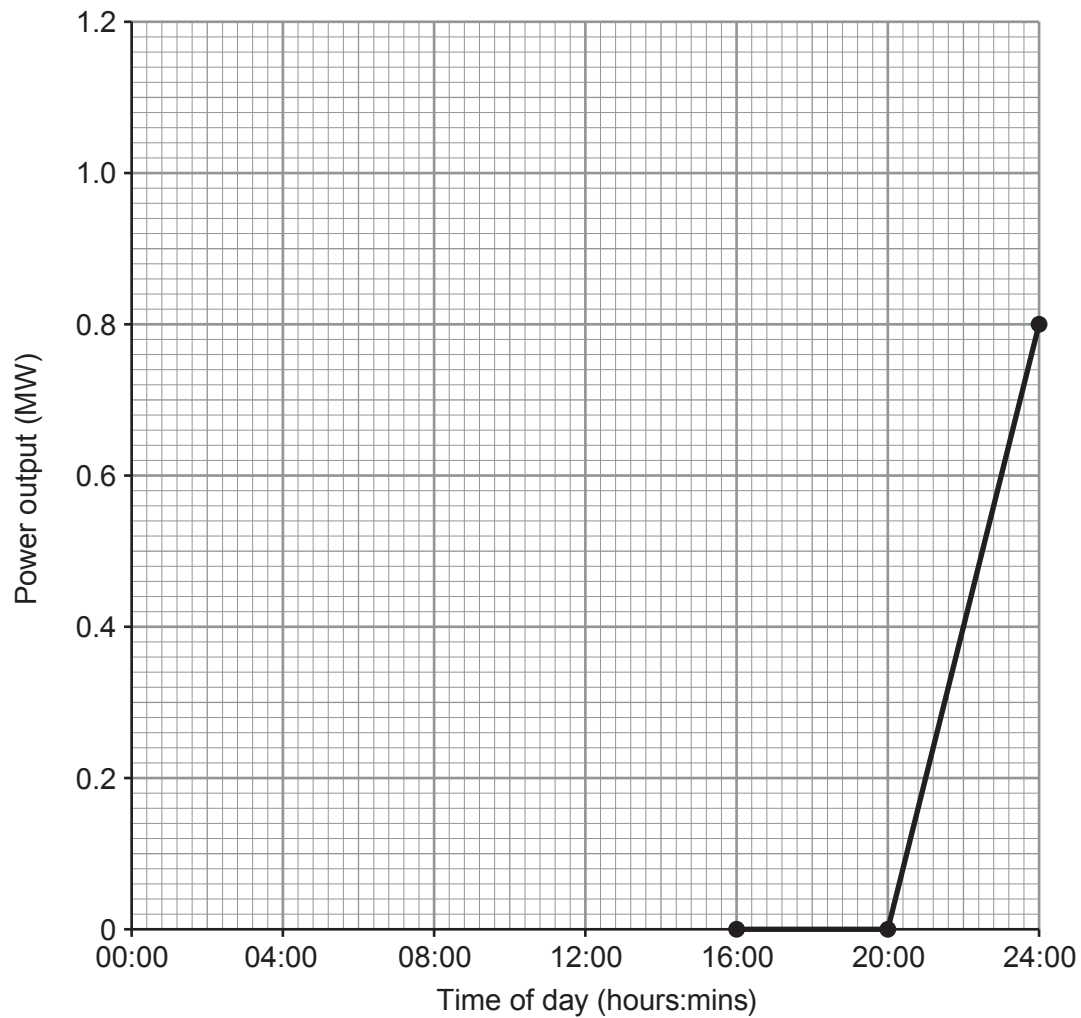
solar

- (b) The information below shows the power output of a wind turbine in one day.

Time of day (hours:mins)	Power output of wind turbine (MW)
00:00	0.6
04:00	1.0
08:00	1.2
12:00	0.8
16:00	0.0
20:00	0.0
24:00	0.8



- (i) Use the data from the table to complete the graph on the grid below and join the points with a ruler. The last three points have been plotted for you. [3]



- (ii) State why the power generated by this wind turbine is not constant during this day. [1]

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(iii) Use the information to answer the following questions.

I. State the maximum power produced by the wind turbine during this day. [1]

maximum power = MW

II. State the time of day that the wind speed was greatest. [1]

time =

III. Determine between which hours the wind turbine did not generate any electricity. [1]

from: to:

(iv) David suggested that the wind turbine generated a power greater than 0.6 MW for 16 hours.

Add a horizontal line to your graph and use it to explain whether David was correct. [4]

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(c) Last year this wind turbine had a mean power output of 0.6 MW. A typical nuclear power station has a continuous power output of 3600 MW.

Calculate how many wind turbines would be needed to generate the same power output as a nuclear power station. [2]

number of wind turbines =

